MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DECODER, DIGITAL, KEY CODE MFP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic decoder module.

1.2 Classification. This specification covers two classes of modules.

Class I and Class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:49Z
Check the source to verify that this is the currentversion before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0921 failures per million operating hours for class I or 0.1658 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MFP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/332-1 for class I and M28787/332-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

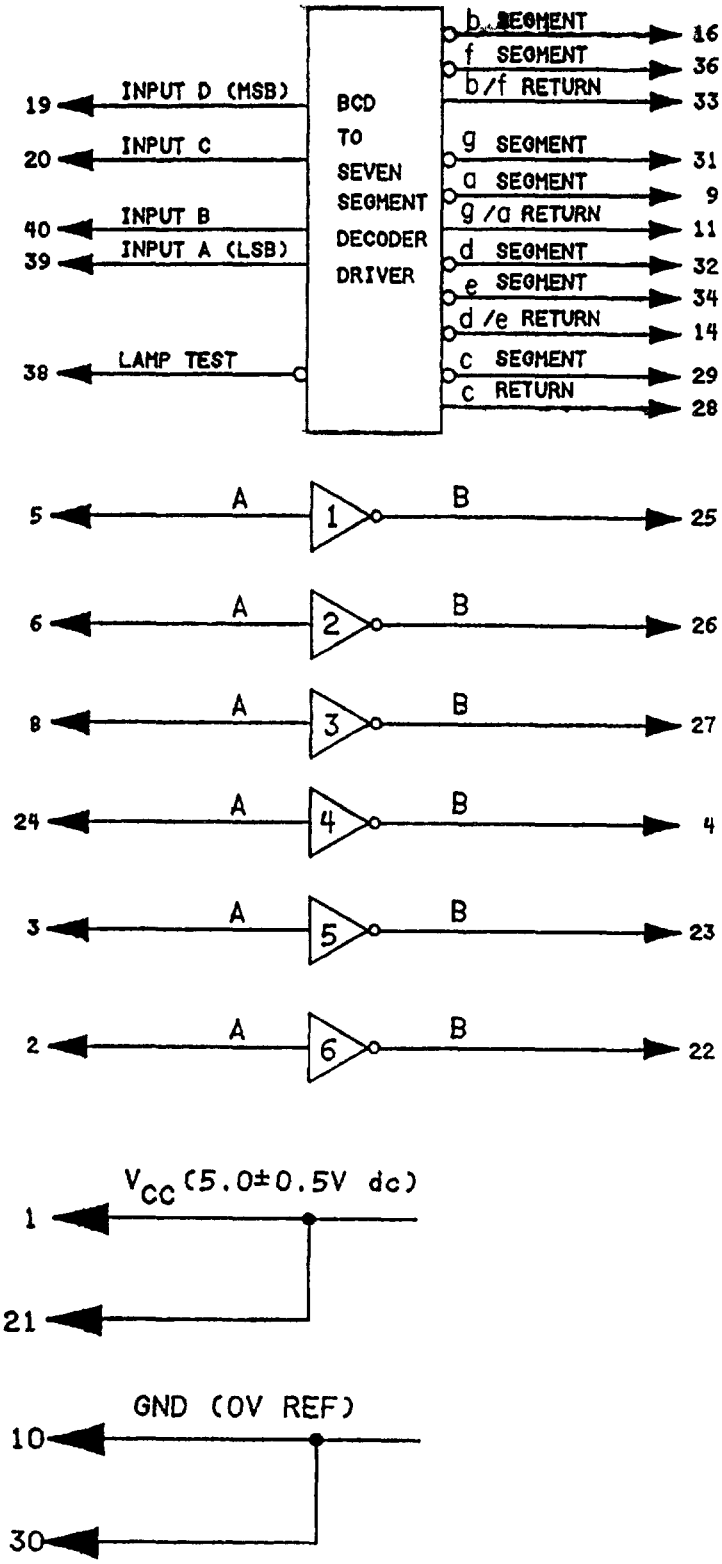
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: 7, 12, 13, 15, 17, 18, 35, and 37 are unused pins.

FIGURE 1. MFP module function diagram, decoder.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
A2	Isolation													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
A3-A9	Isolation	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
B1,B2	Continuity	IV	X	0	1	0	1	0	1	0	1	0	1	Ω
C	V_{CC} stress	V	S	← Module must pass all remaining tests →										
D1-D6	Input current, low level	VI	X	0.05	0.40	0.05	0.40	0.05	0.40	0.05	0.40	0.05	0.40	mA
D7-D11	Input current, low level	VI	X	0.2	1.6	0.2	1.6	0.2	1.6	0.2	1.6	0.2	1.6	mA
E1-E6	Input current, high level	VII	X	0	20	0	20	0	20	0	20	0	20	µA
E7-E11	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
F1-F6	Output short circuit current	VIII	X	20	100	20	100	20	100	20	100	20	100	mA
G1-G6	Input breakdown current	IX	S	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	mA
G7-G11	Input breakdown current	IX	S	0	1	0	1	0	1	0	1	0	1	mA
H1-H12	Functional	X	X	← Test vectors must produce output vectors as specified in table X →										
J1-J17	Functional	XI	X	← Test vectors must produce output vectors as specified in table XI →										
K1-K6	Output voltage, low level	XII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L6	Output voltage, high level	XIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
M1-M7	Off state collector leakage current	XIV	X	0	50	0	50	0	50	0	50	0	50	nA
N1-N7	Output voltage (ON)	XV	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
P	Power supply current	XVI	X	10	150	10	150	10	150	10	150	10	150	mA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R6	Propagation delay time (t_{PHL}), C_L = 50 pF	XVII	S	2	25	2	25	2	25	2	25	2	25	ns
R7-R12	Propagation delay time (t_{PLH}), C_L = 50 pF	XVII	S	2	25	2	25	2	25	2	25	2	25	ns
S1-S26	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	10	230	10	230	10	230	10	230	10	230	ns
S27-S52	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	10	230	10	230	10	230	10	230	10	230	ns
T1-T6	Output crosstalk, low level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
U1-U6	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
V	Power supply transient amplitude	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
W	Power supply filter capacitance	XXII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
X1-X11	Input capacitance	XXIII	Q	2	30	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, C, G, or F are part of a sequence of tests, the order of testing shall be as follows: A, C, G, F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and D through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B and D through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	550	550	mW
Supply voltage (see 6.5.1)	5.5	5.5	V
Input levels			
Low Level	0.8 maximum	0.8 maximum	V
High level	2 minimum	2 minimum	V

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T17:49Z
Check the source to verify that this is the current version before use.

Decoder source voltage	24	24	V
Decoder sink current	0.1	0.1	A

6.5 Module function. This module converts 4-bit binary data of either polarity to seven segment display data with high drive capability.

6.5.1 Maximum ratings. The maximum ratings for supply voltages and input levels given in 6.4 represent extremes beyond which damage to the module may occur. Operation at the extremes is performed in this specification to ensure that the module will survive these levels without damage. Performance specifications are tested at supply levels 8 percent less than maximum ratings and digital and analog input levels equal to or between the supply voltages.

6.5.2 Input to output relationship. The digital input versus seven segment output relationship is shown in table II.

Agent:
DLA - ES

Preparing activity:
Navy - EC

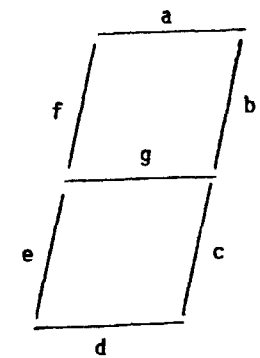
(Project 5963-N463)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE II. Input to output relationship. 1/ - 5/

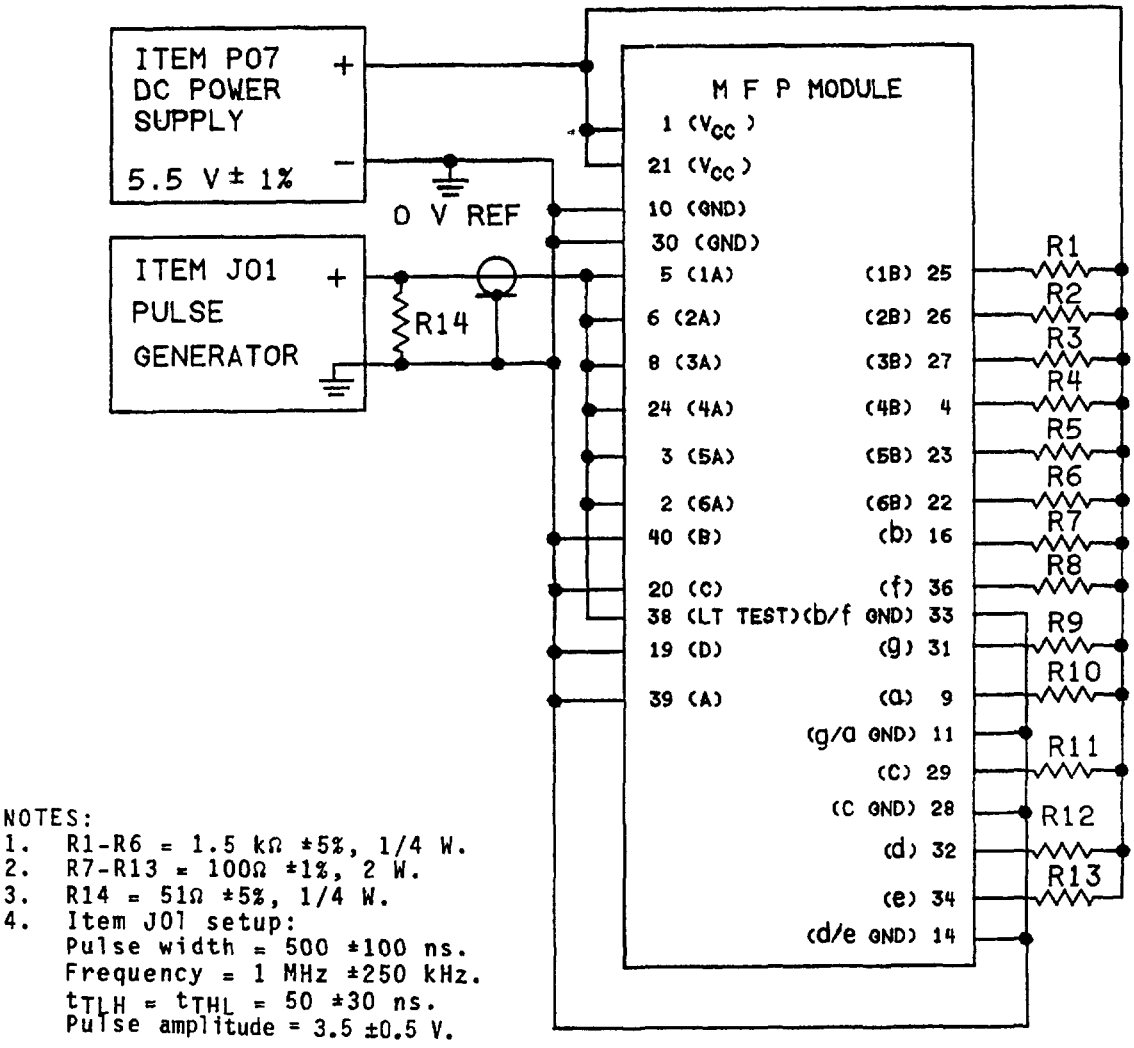
Decimal or function	Inputs					Outputs						
	LT	D	C	B	A	a	b	c	d	e	f	g
0	H	L	L	L	L	ON	ON	ON	ON	ON	ON	OFF
1	H	L	L	L	H	OFF	ON	ON	OFF	OFF	OFF	OFF
2	H	L	L	H	L	ON	ON	OFF	ON	ON	OFF	ON
3	H	L	L	H	H	ON	ON	ON	ON	OFF	OFF	ON
4	H	L	H	L	L	OFF	ON	ON	OFF	OFF	ON	ON
5	H	L	H	L	H	ON	OFF	ON	ON	OFF	ON	ON
6	H	L	H	H	L	OFF	OFF	ON	ON	ON	ON	ON
7	H	L	H	H	H	ON	ON	ON	OFF	OFF	OFF	OFF
8	H	H	L	L	L	ON	ON	ON	ON	ON	ON	ON
9	H	H	L	L	H	ON	ON	ON	OFF	OFF	ON	ON
10	H	H	L	H	L	OFF	OFF	OFF	ON	ON	OFF	ON
11	H	H	L	H	H	OFF	OFF	ON	ON	OFF	OFF	ON
12	H	H	H	L	L	OFF	ON	OFF	OFF	OFF	ON	ON
13	H	H	H	L	H	ON	OFF	OFF	ON	OFF	ON	ON
14	H	H	H	H	L	OFF	OFF	OFF	ON	ON	ON	ON
15	H	H	H	H	H	OFF	OFF	OFF	OFF	OFF	OFF	OFF
LT	L	X	X	X	X	ON	ON	ON	ON	ON	ON	ON

1/ H = High voltage level.
2/ L = Low voltage level.
3/ X = Irrelevant.
4/ ON = Transistor on.
5/ OFF = Transistor off.



Seven segment display

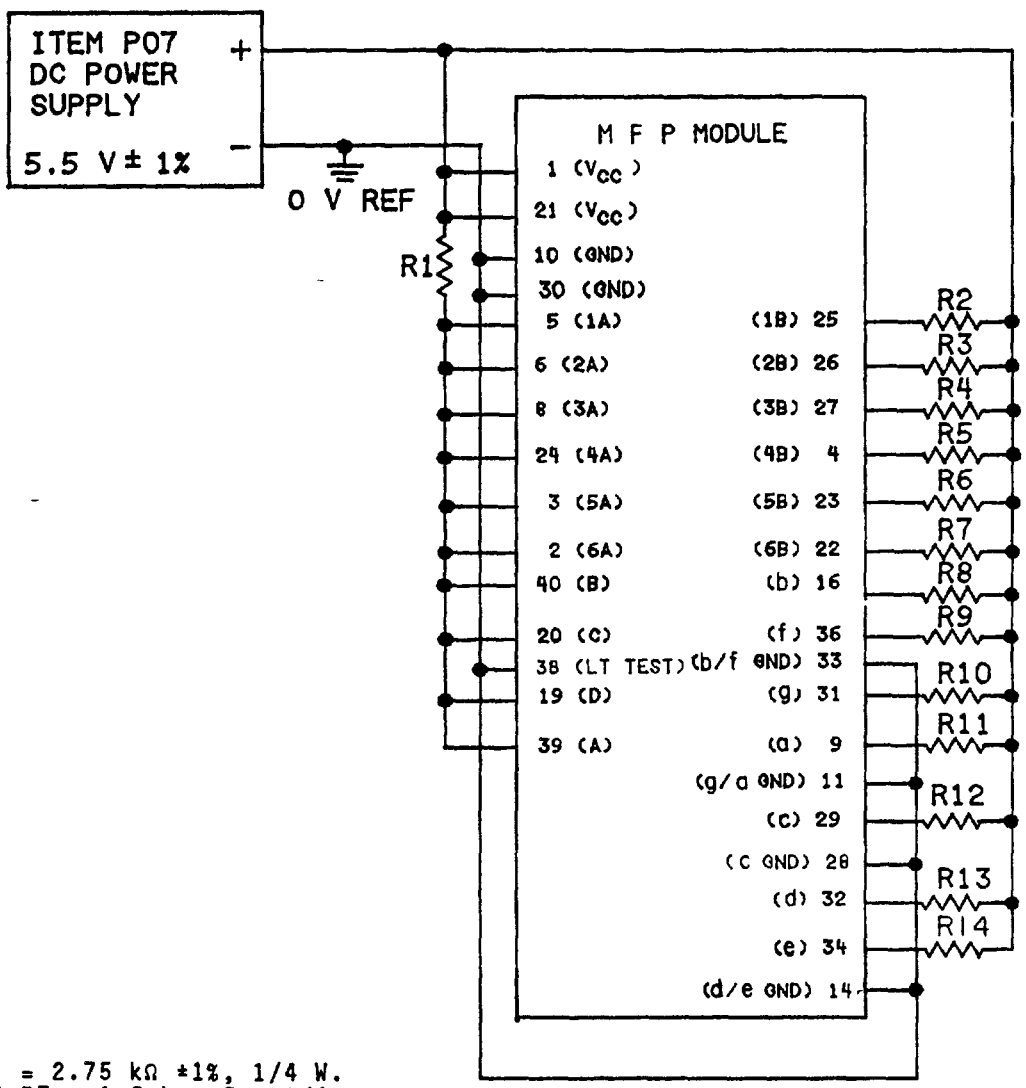
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R6 = 1.5 kΩ ±5%, 1/4 W.
2. R7-R13 = 100Ω ±1%, 2 W.
3. R14 = 51Ω ±5%, 1/4 W.
4. Item J01 setup:
   Pulse width = 500 ±100 ns.
   Frequency = 1 MHz ±250 kHz.
   $t_{TLH} = t_{THL}$ = 50 ±30 ns.
   Pulse amplitude = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 2.75 kΩ ±1%, 1/4 W.
2. R2-R7 = 1.5 kΩ ±5%, 1/4 W.
3. R8-R14 = 39Ω ±5%, 2 W.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

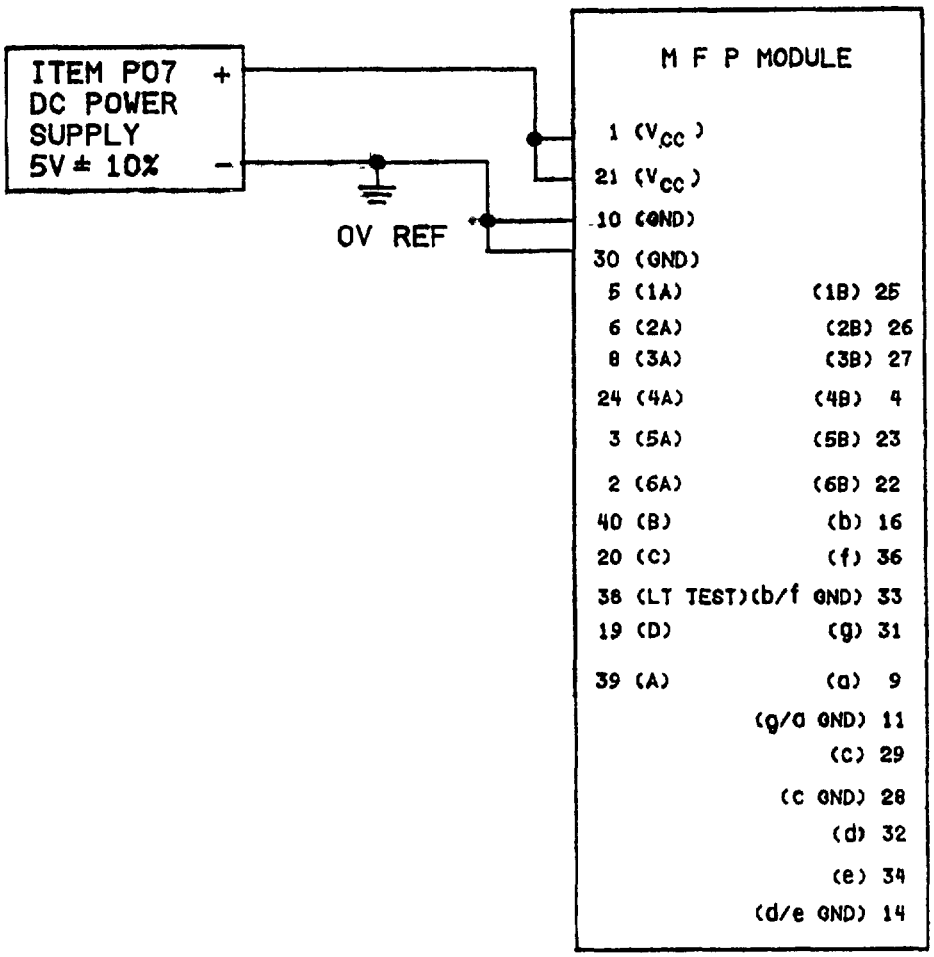


FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

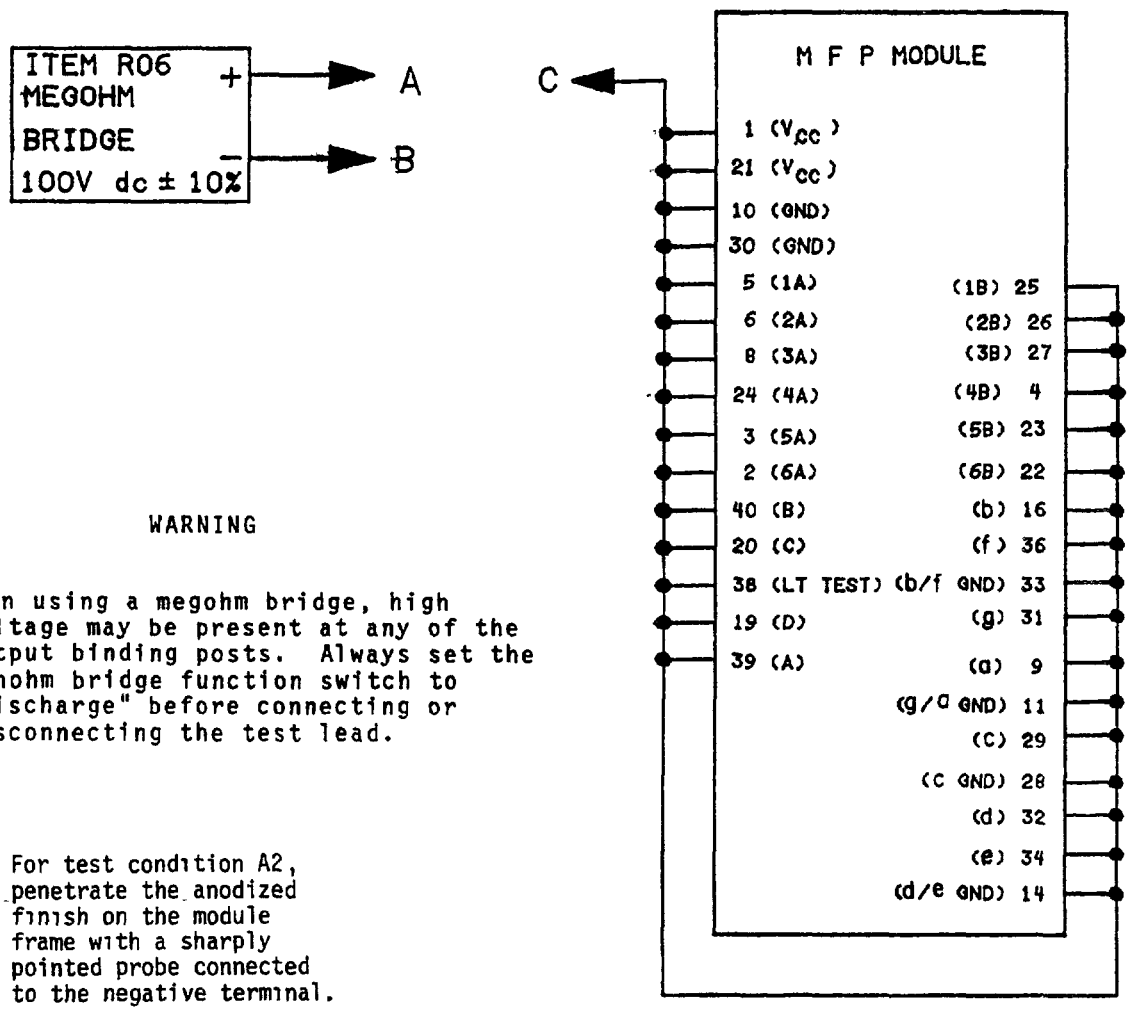
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C	7,12,13,15,17,18,35,37	10	∞	MΩ	5
A2	"	C,7,12,13,15, 17,18,35,37	Frame	"	"	"	"
A3	"	7	12,13,15,17,18,35,37	"	"	"	"
A4	"	12	13,15,17,18,35,37	"	"	"	"
A5	"	13	15,17,18,35,37	"	"	"	"
A6	"	15	17,18,35,37	"	"	"	"
A7	"	17	18,35,37	"	"	"	"
A8	"	18	35,37	"	"	"	"
A9	"	35	37	"	"	"	"

TABLE IV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Continuity	1	21	0	1	Ω	6
B2	Continuity	10	30	0	1	Ω	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the mehohm bridge function switch to "discharge" before connecting or disconnecting the test lead.

NOTE: For test condition A2, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:49Z
Check the source to verify that this is the current version before use.

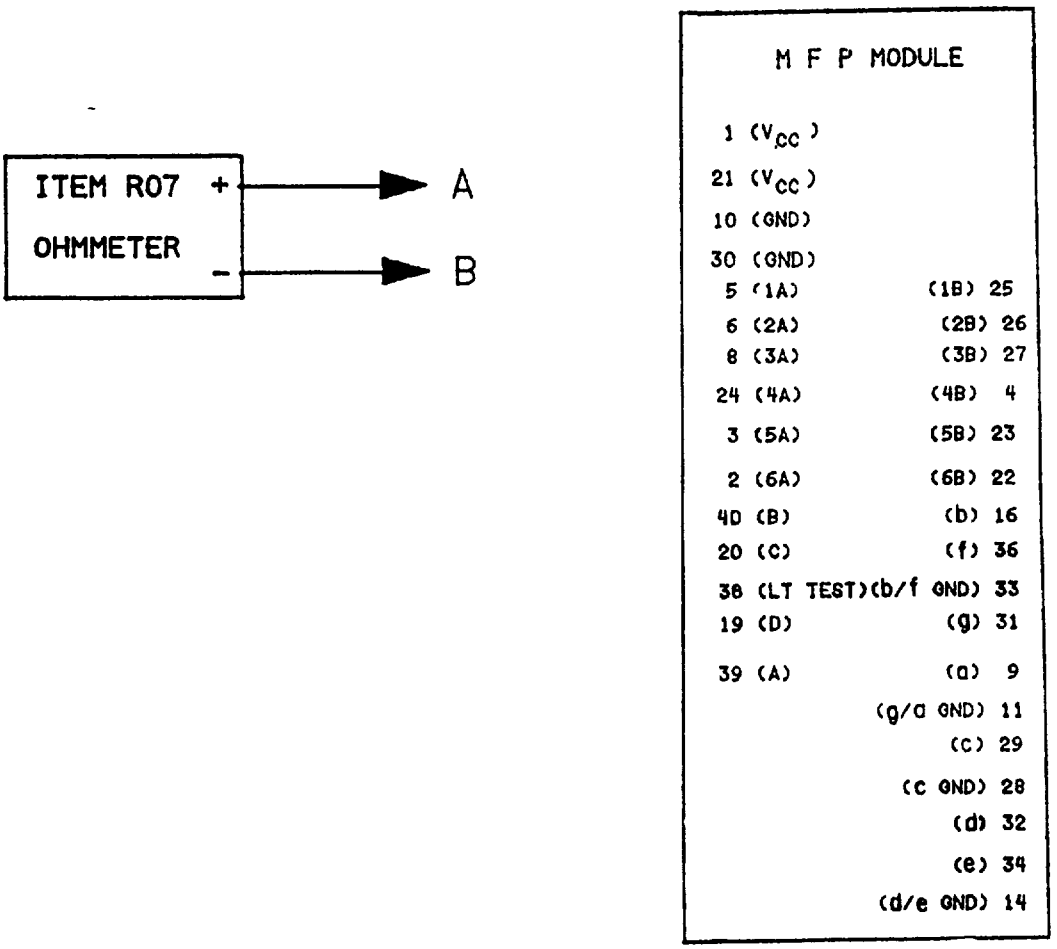


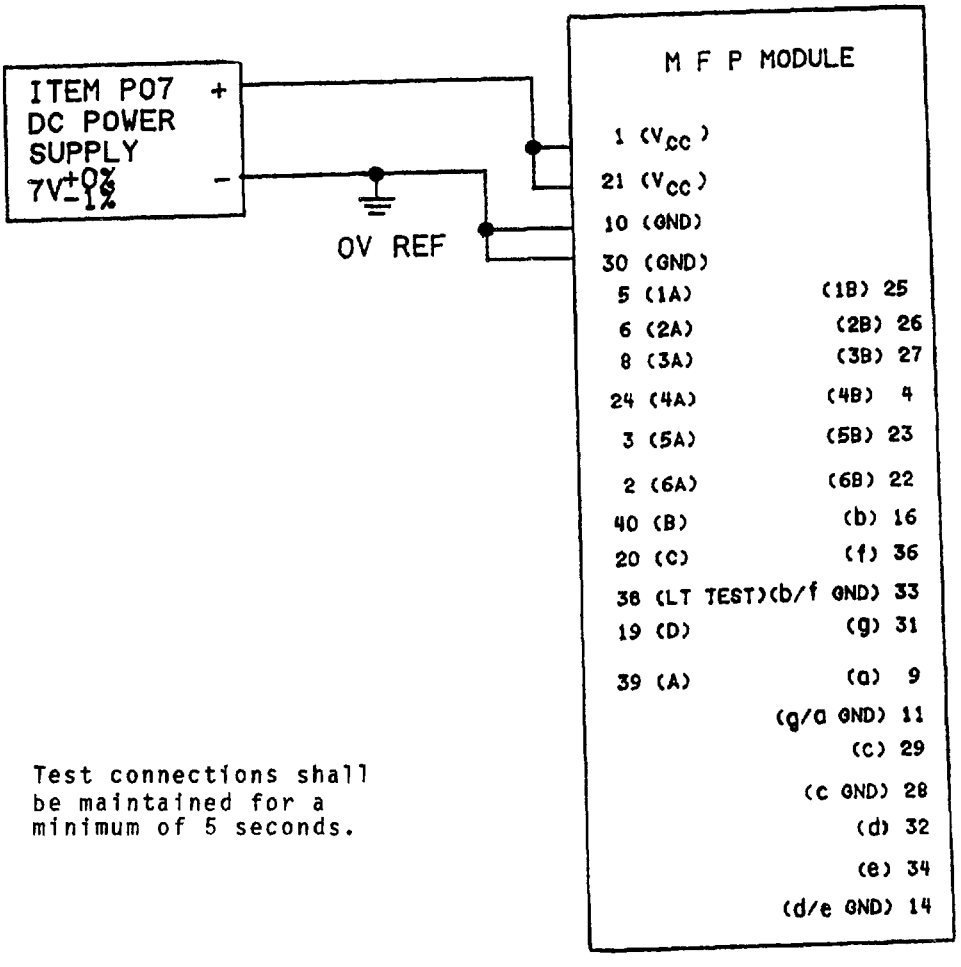
FIGURE 6. Continuity test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE V. V_{CC} stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C	V_{CC} stress	Module must pass all remaining tests.			7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: Test connections shall be maintained for a minimum of 5 seconds.

FIGURE 7. V_{CC} stress test for table V.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE VI. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Input current, low level	Open	5	Open	0.05	0.40	mA	8
D2	"	"	6	"	"	"	"	"
D3	"	"	8	"	"	"	"	"
D4	"	"	24	"	"	"	"	"
D5	"	"	3	"	"	"	"	"
D6	"	"	2	"	"	"	"	"
D7	"	"	40	"	0.2	1.6	"	"
D8	"	"	20	"	"	"	"	"
D9	"	"	38	"	"	"	"	"
D10	"	"	19	"	"	"	"	"
D11	"	"	39	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current (high-level) test.

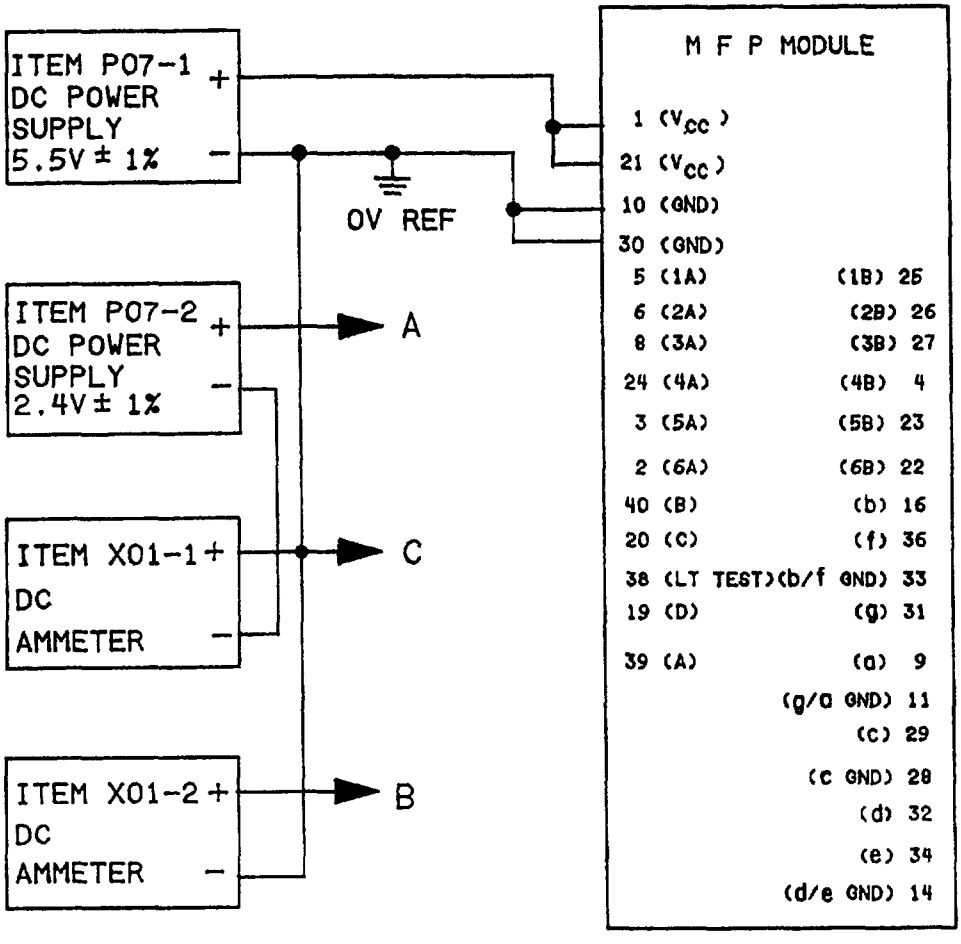
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	Input current, high level	5	Open	Open	0	20	µA	8
E2	"	6	"	"	"	"	"	"
E3	"	8	"	"	"	"	"	"
E4	"	24	"	"	"	"	"	"
E5	"	3	"	"	"	"	"	"
E6	"	2	"	"	"	"	"	"
E7	"	40	"	"	"	40	"	"
E8	"	20	"	"	"	"	"	"
E9	"	38	"	"	"	"	"	"
E10	"	19	"	"	"	"	"	"
E11	"	39	"	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	Output short circuit current	Open	25	5	20	100	mA	8
F2	"	"	26	6	"	"	"	"
F3	"	"	27	8	"	"	"	"
F4	"	"	4	24	"	"	"	"
F5	"	"	23	3	"	"	"	"
F6	"	"	22	2	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:49Z
Check the source to verify that this is the current version before use.



CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables VI and VIII and item X01-1 for table VII.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 μA range.
3. When performing output short circuit current tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

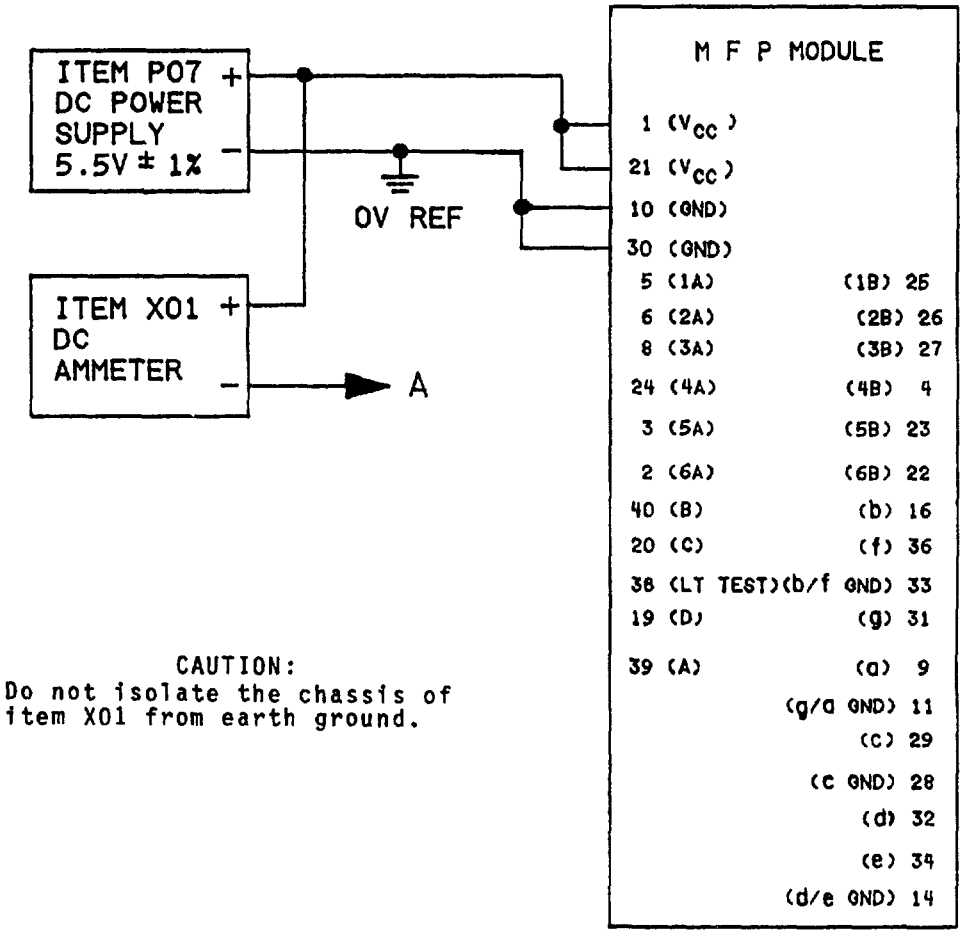
FIGURE 8. Input current (low-level and high-level) and output short circuit current test circuit for tables VI, VII, and VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Input breakdown current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	Input breakdown current	5	0.0	0.1	mA	9
G2	"	6	"	"	"	"
G3	"	8	"	"	"	"
G4	"	24	"	"	"	"
G5	"	3	"	"	"	"
G6	"	2	"	"	"	"
G7	"	40	0	1	"	"
G8	"	20	"	"	"	"
G9	"	38	"	"	"	"
G10	"	19	"	"	"	"
G11	"	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



CAUTION:
Do not isolate the chassis of item X01 from earth ground.

FIGURE 9. Input breakdown current test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE X. Functional test.

Test condition	Parameter	Connect			Output logic status 1/	Test figure
		A to pin	B to pin	C to pin		
H1	Functional	5	Open	25	L	10
H2	"	6	"	26	L	"
H3	"	8	"	27	L	"
H4	"	24	"	4	L	"
H5	"	3	"	23	L	"
H6	"	2	"	22	L	"
H7	"	Open	5	25	H	"
H8	"	"	6	26	H	"
H9	"	"	8	27	H	"
H10	"	"	24	4	H	"
H11	"	"	3	23	H	"
H12	"	"	2	22	H	"

1/ Output logic status:
L = Low voltage level = 0.0 V to 0.5 V.
H = High voltage level = 2.5 V to 5.0 V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

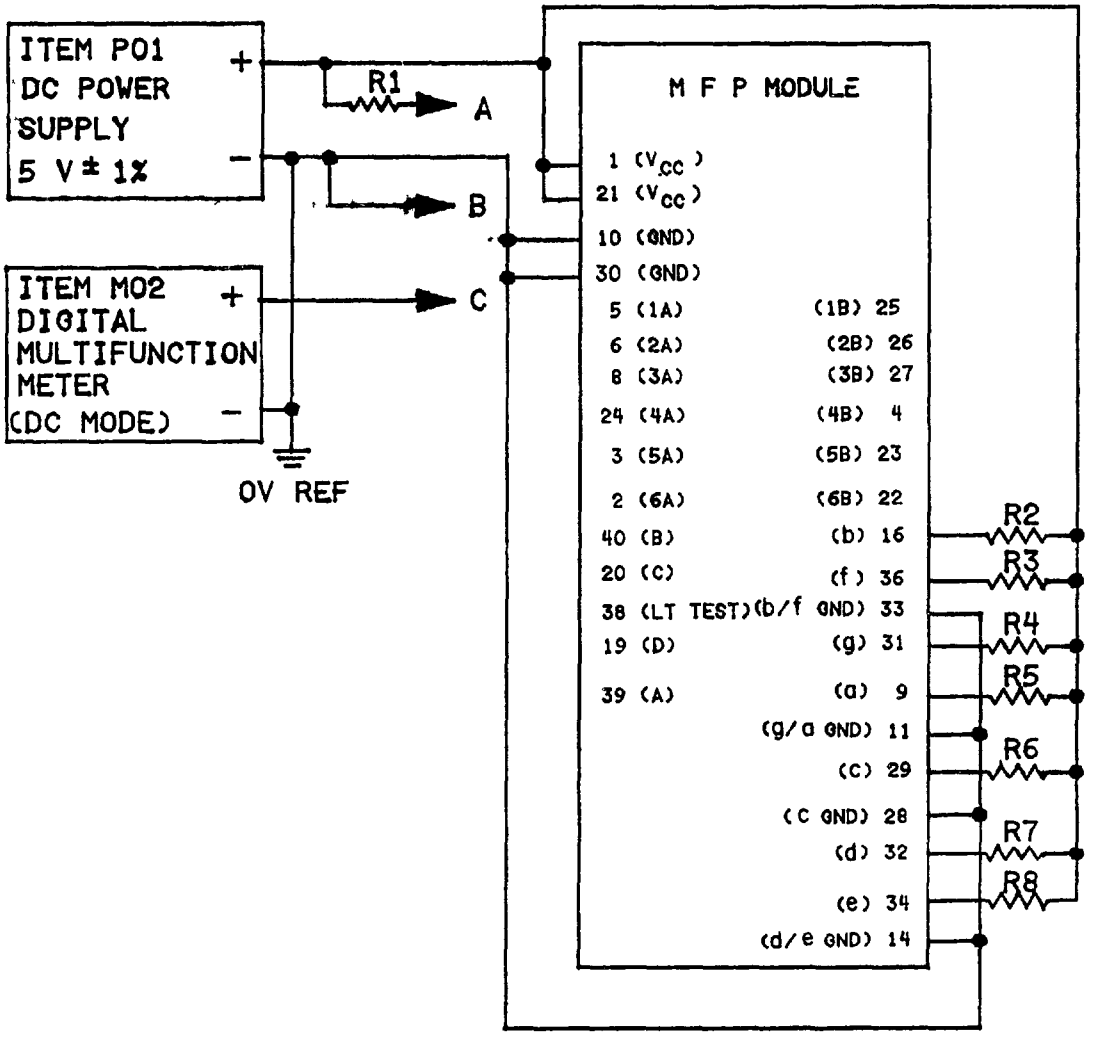
TABLE XI. Functional Test.

Test condition	Parameter	Connect			Output logic status 1/							Test figure
		A to pin	B to pin	C to pin	a	b	c	d	e	f	g	
J1	Functiona'	Open	38	2/	L	L	L	L	L	L	L	10
J2	"	38	39,40,20,19	"	L	L	L	L	L	L	H	"
J3	"	38,39	40,20,19	"	H	L	L	H	H	H	H	"
J4	"	38,40	39,20,19	"	L	L	H	L	L	H	L	"
J5	"	38,39,40	20,19	"	L	L	L	L	H	H	L	"
J6	"	38,20	39,40,19	"	H	L	L	H	H	L	L	"
J7	"	38,39,20	40,19	"	L	H	L	L	H	L	L	"
J8	"	38,40,20	39,19	"	H	H	L	L	L	L	L	"
J9	"	38,39,40,20	19	"	L	L	L	H	H	H	H	"
J10	"	38,19	39,40,20	"	L	L	L	L	L	L	L	"
J11	"	38,39,19	40,20	"	L	L	L	H	H	L	L	"
J12	"	38,40,19	39,20	"	H	H	H	L	L	H	L	"
J13	"	38,39,40,19	20	"	H	H	L	L	H	H	L	"
J14	"	38,20,19	39,40	"	H	L	H	H	H	L	L	"
J15	"	38,39,20,19	40	"	L	H	H	L	H	L	L	"
J16	"	38,40,20,19	39	"	H	H	H	L	L	L	L	"
J17	"	38,39,40, 20,19	Open	"	H	H	H	H	H	H	H	"

1/ Output logic status:
L = Low voltage level = 0.0 V to 0.5 V.
H = High voltage level = 4.5 V to 5.1 V.

2/ For test conditions J1-J16, verification of the output logic status in footnote 1 is obtained by connecting item MO2, point C, to each output in succession after input conditions are accomplished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: R1-R8 = 2 kΩ ±1%, 1/4 W.

FIGURE 10. Functional test circuit for tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:49Z
Check the source to verify that this is the current version before use.

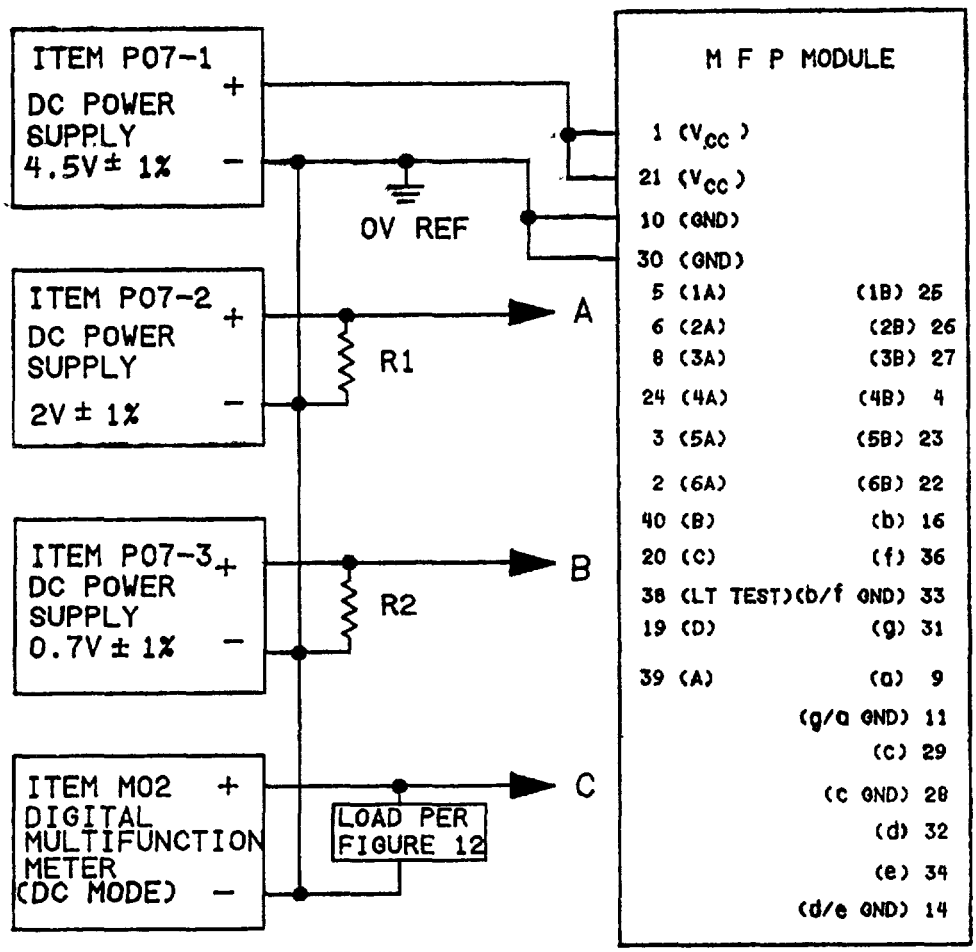
TABLE XII. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Output voltage, low level	5	Open	25	0.0	0.4	V	11
K2	"	6	"	26	"	"	"	"
K3	"	8	"	27	"	"	"	"
K4	"	24	"	4	"	"	"	"
K5	"	3	"	23	"	"	"	"
K6	"	2	"	22	"	"	"	"

TABLE XIII. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Output voltage, high level	Open	5	25	2.5	4.5	V	11
L2	"	"	6	26	"	"	"	"
L3	"	"	8	27	"	"	"	"
L4	"	"	24	4	"	"	"	"
L5	"	"	3	23	"	"	"	"
L6	"	"	2	22	"	"	"	"

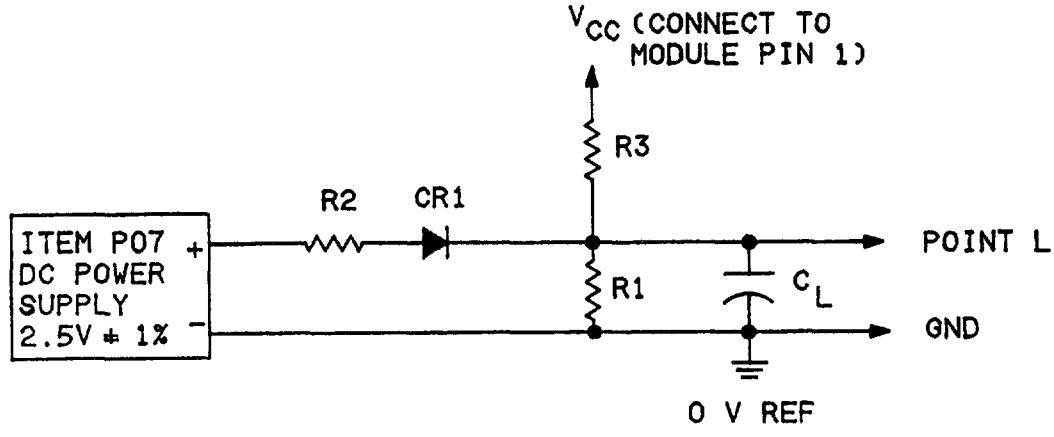
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 11. Output voltage (low-level and high-level) test circuit for tables XII and XIII.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 6.04 kΩ ±1%, 1/8 W with a current flow of -400 µA ±5% out of point L with point L held at 2.4 V.
3. R2 = Select value for a current flow of 4 mA ±5% out of point L with point L held at 0.4 V.
4. R3 = 2 kΩ ±1%, 1/4 W (used only for test conditions).
5. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions R, S, T, and U.
6. C_L is not required for test conditions K and L.

FIGURE 12. MFP logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

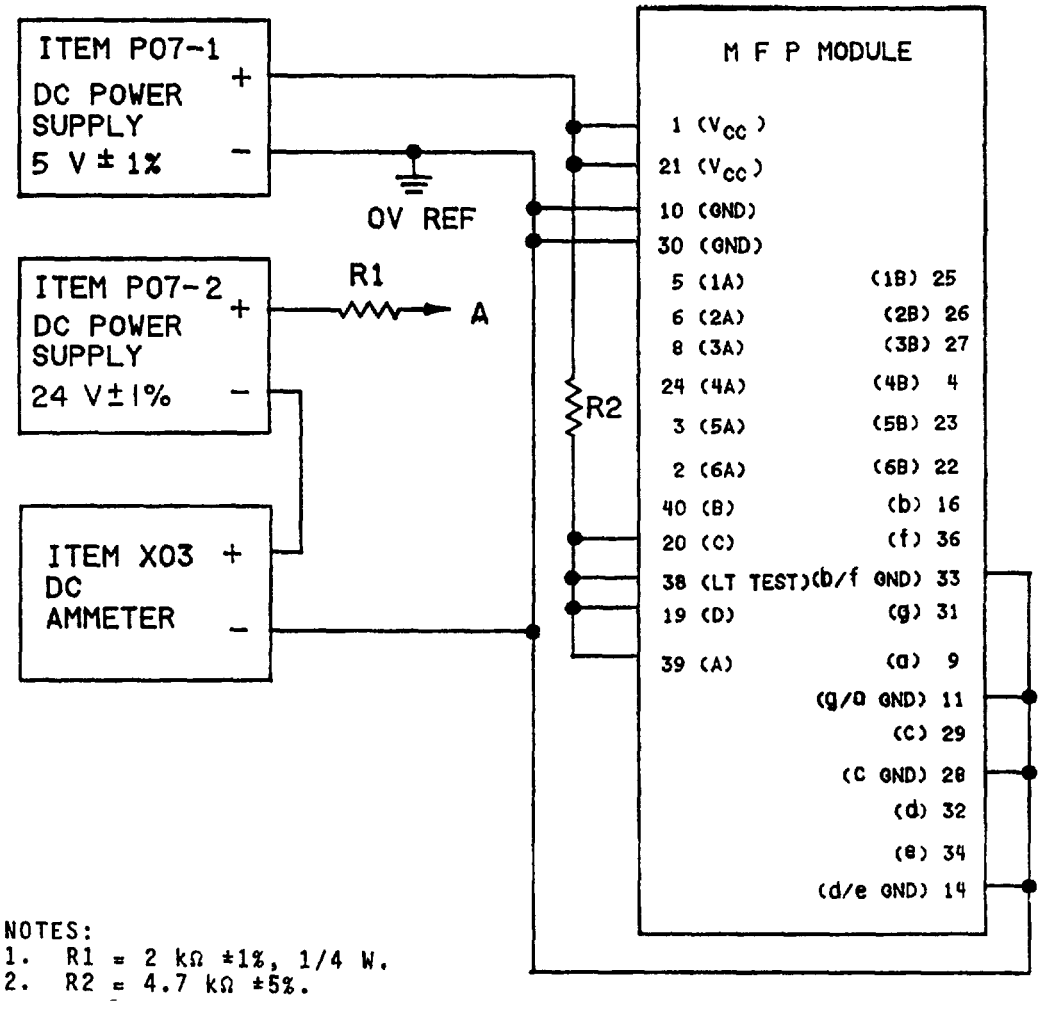
TABLE XIV. Off state collector leakage current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
M1	Off state collector leakage current	16	0	50	nA	13
M2	"	36	"	"	"	"
M3	"	31	"	"	"	"
M4	"	9	"	"	"	"
M5	"	29	"	"	"	"
M6	"	32	"	"	"	"
M7	"	34	"	"	"	"

TABLE XV. Output voltage (ON) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Output voltage (ON)	16	16	0.0	0.4	V	14
N2	"	36	36	"	"	"	"
N3	"	31	31	"	"	"	"
N4	"	9	9	"	"	"	"
N5	"	29	29	"	"	"	"
N6	"	32	32	"	"	"	"
N7	"	34	34	"	"	"	"

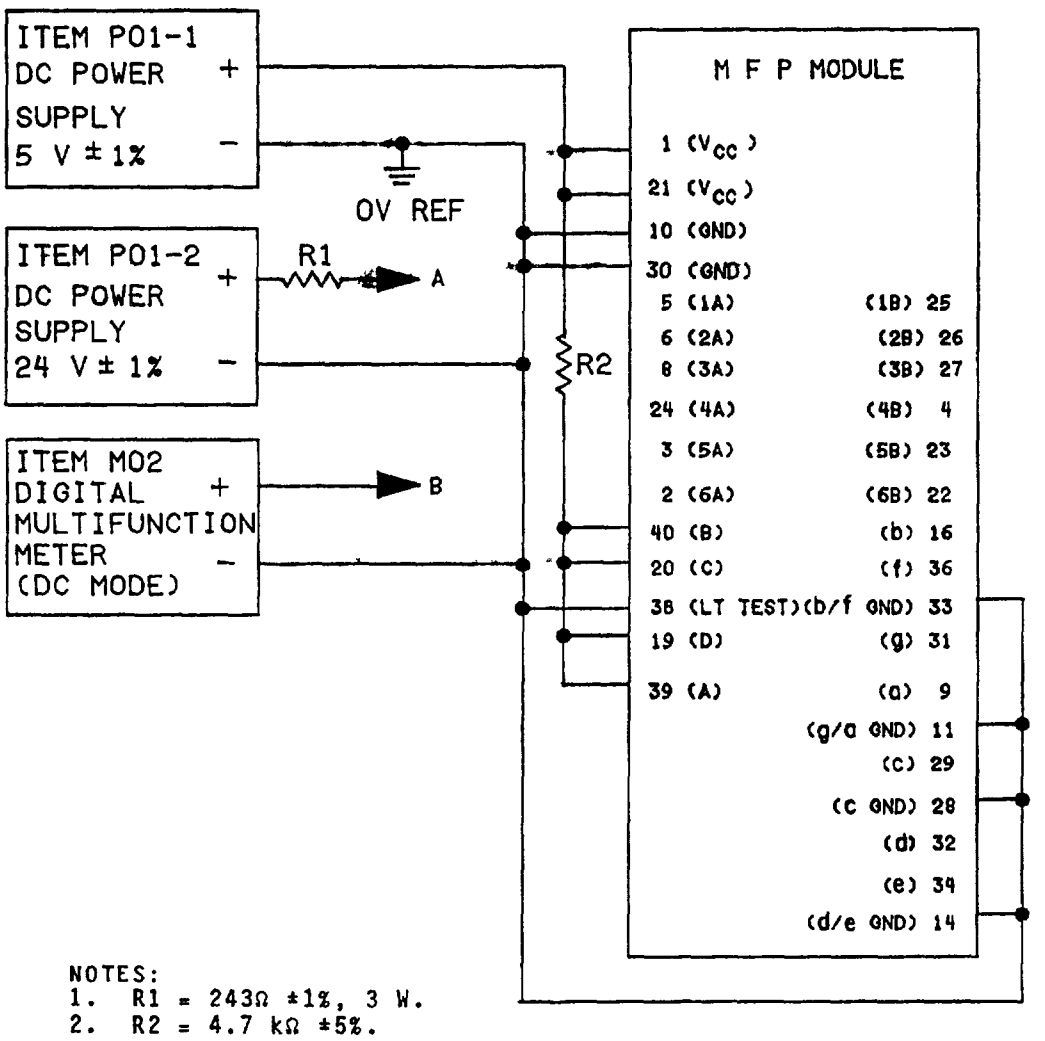
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 2 kΩ ±1%, 1/4 W.
2. R2 = 4.7 kΩ ±5%.

FIGURE 13. Off state collector leakage current test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 243Ω ±1%, 3 W.
2. R2 = 4.7 kΩ ±5%.

FIGURE 14. Output voltage (ON) test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE XVI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply current	10	150	mA	15

TABLE XVII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Propagation delay time (t_{PHL}), C_L = 50 pF	5	25	2	25	ns	16,17
R2	"	6	26	"	"	"	"
R3	"	8	27	"	"	"	"
R4	"	24	4	"	"	"	"
R5	"	3	23	"	"	"	"
R6	"	2	22	"	"	"	"
R7	Propagation delay time (t_{PLH}), C_L = 50 pF	5	25	"	"	"	"
R8	"	6	26	"	"	"	"
R9	"	8	27	"	"	"	"
R10	"	24	4	"	"	"	"
R11	"	3	23	"	"	"	"
R12	"	2	22	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

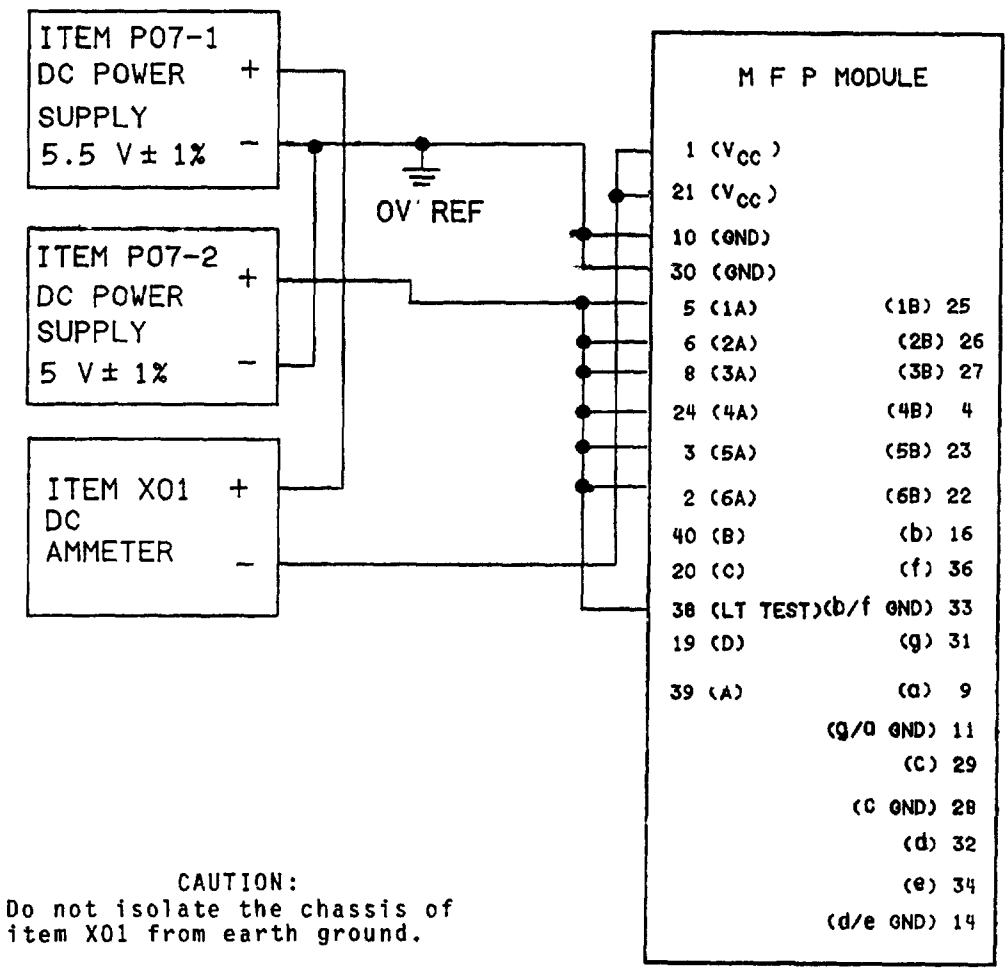
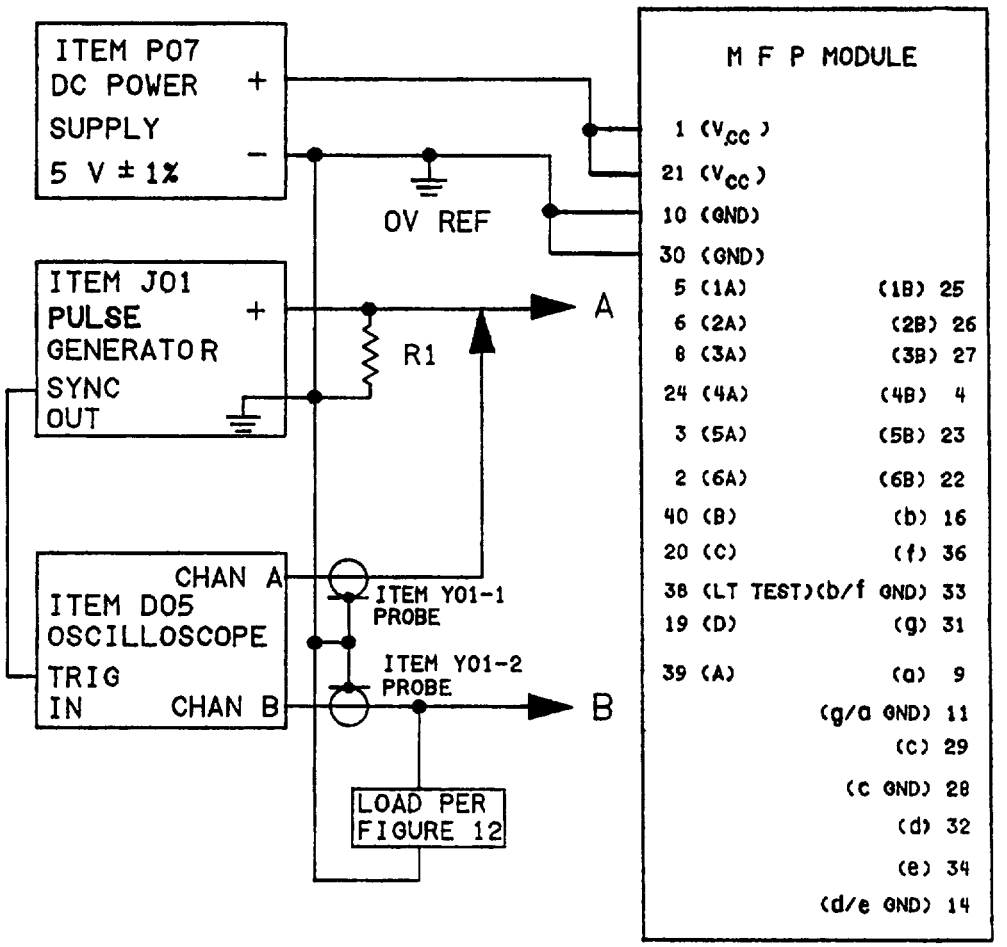


FIGURE 15. Power supply current test circuit for table XVI.

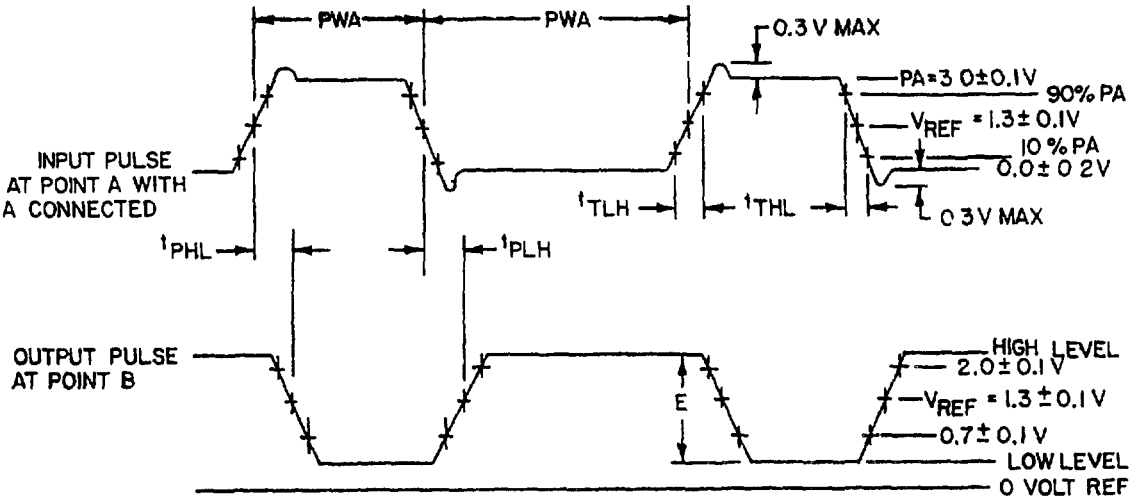
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 16. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
2. Before making the measurements specified in table XVII, verify that the output pulse high level is $\geq$ 2.4 and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:49Z
Check the source to verify that this is the currentversion before use.

TABLE XVIII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	1/	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Propagation delay time (t_{PHL}), C_L = 50 pF	-	Open	40,20,19	39	9	10	230	ns	18,19
S2	"	-	"	"	"	32	"	"	"	"
S3	"	-	"	"	"	34	"	"	"	"
S4	"	-	"	"	"	36	"	"	"	"
S5	"	+	40,20	19	"	16	"	"	"	"
S6	"	-	40,20	19	"	31	"	"	"	"
S7	"	+	40	20,19	"	29	"	"	"	"
S8	"	-	19	39,20	40	9	"	"	"	"
S9	"	-	"	"	"	16	"	"	"	"
S10	"	-	"	"	"	29	"	"	"	"
S11	"	-	"	"	"	36	"	"	"	"
S12	"	-	39,20	19	"	31	"	"	"	"
S13	"	-	39,20	19	"	32	"	"	"	"
S14	"	+	20,19	39	"	34	"	"	"	"
S15	"	+	39	40,19	20	9	"	"	"	"
S16	"	+	"	"	"	32	"	"	"	"
S17	"	+	"	"	"	36	"	"	"	"
S18	"	+	"	"	"	31	"	"	"	"
S19	"	-	39,19	40	"	16	"	"	"	"
S20	"	-	39,19	40	"	29	"	"	"	"
S21	"	-	19	39,40	"	34	"	"	"	"

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	1/	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S22	Propagation delay time (t_{PHL}), C_L = 50 pF	-	39,40,20	Open	19	9	10	230	ns	18,19
S23	"	-	"	"	"	16	"	"	"	"
S24	"	-	"	"	"	29	"	"	"	"
S25	"	+	39	40,20	"	31	"	"	"	"
S26	"	+	39	40,20	"	36	"	"	"	"
S27	Propagation delay time (t_{PLH}), C_L = 50 pF	+	Open	40,29,19	39	9	"	"	"	"
S28	"	+	"	"	"	32	"	"	"	"
S29	"	+	"	"	"	34	"	"	"	"
S30	"	+	"	"	"	36	"	"	"	"
S31	"	-	40,20	19	"	16	"	"	"	"
S32	"	+	40,20	19	"	31	"	"	"	"
S33	"	-	40	20,19	"	29	"	"	"	"
S34	"	+	19	39,20	40	9	"	"	"	"
S35	"	+	"	"	"	16	"	"	"	"
S36	"	+	"	"	"	29	"	"	"	"
S37	"	+	"	"	"	36	"	"	"	"
S38	"	+	39,20	19	"	31	"	"	"	"
S39	"	+	39,20	19	"	32	"	"	"	"
S40	"	-	20,19	39	"	34	"	"	"	"
S41	"	-	39	40,19	20	9	"	"	"	"

See footnote at end of table.

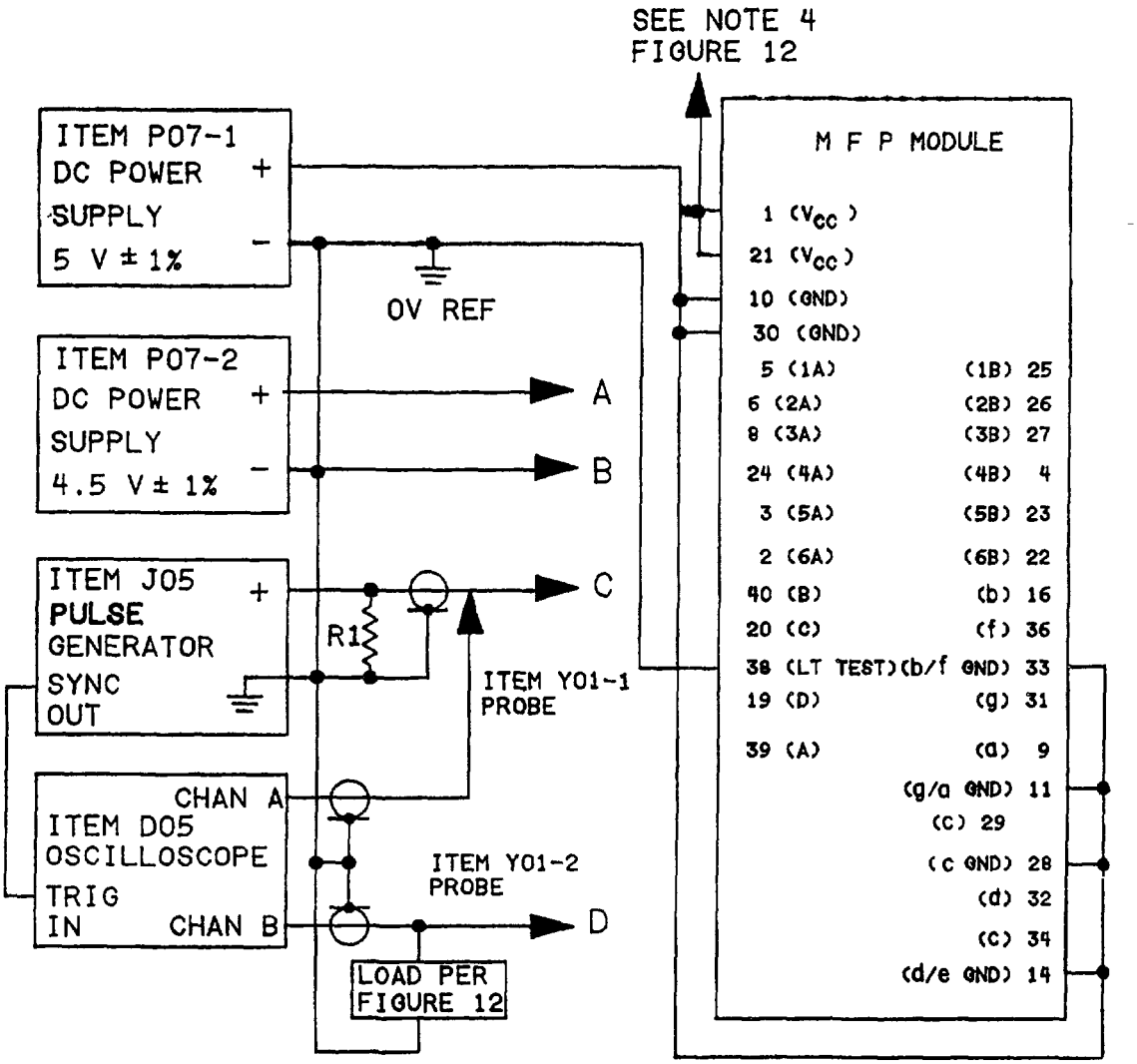
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	1/	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S42	Propagation delay time (t_{PLH}), C_L = 50 pF	-	39	40,19	20	32	10	230	ns	18,19
S43	"	-	"	"	"	36	"	"	"	"
S44	"	-	"	"	"	31	"	"	"	"
S45	"	+	39,19	40	"	16	"	"	"	"
S46	"	+	39,19	40	"	29	"	"	"	"
S47	"	+	19	39,40	"	34	"	"	"	"
S48	"	+	39,40,20	Open	19	9	"	"	"	"
S49	"	+	"	"	"	16	"	"	"	"
S50	"	+	"	"	"	29	"	"	"	"
S51	"	-	39	40,20	"	31	"	"	"	"
S52	"	+	39	40,20	"	36	"	"	"	"

1/ Oscilloscope trigger (see note 4, figure 19).

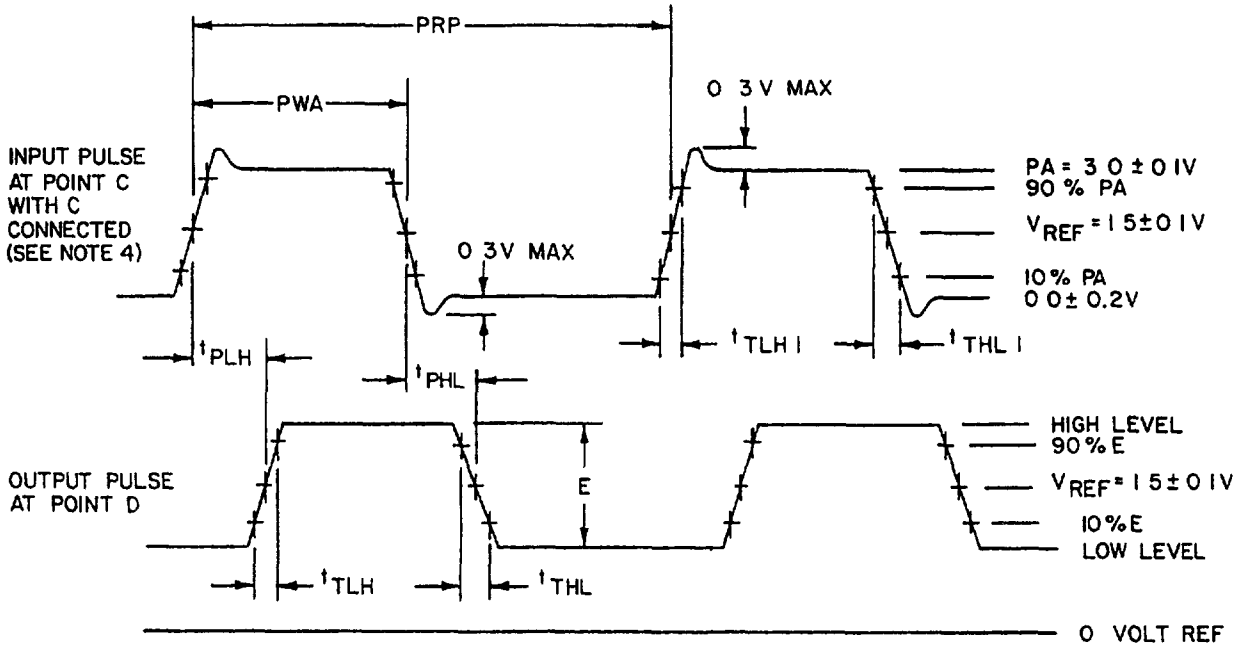
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 18. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in table XVIII, verify that the output high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.
4. When oscilloscope is triggered on negative (-) edge, invert the input waveform.

FIGURE 19. Propagation delay time (t_{PHL} and t_{PLH}) test ciruit waveforms for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

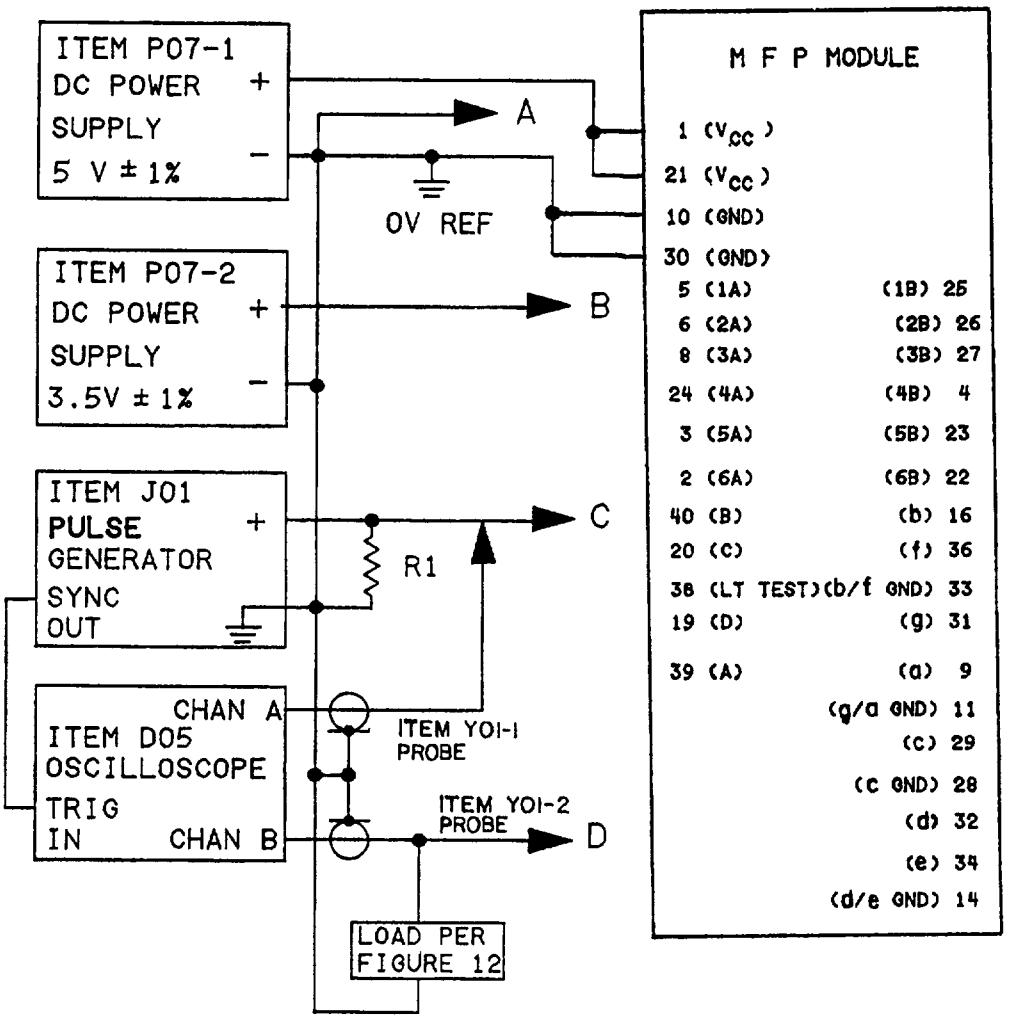
TABLE XIX. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	Output crosstalk, low level	Open	5	6,8,24,3,2,38	25	0	500	mV	20,21
T2	"	"	6	5,8,24,3,2,38	26	"	"	"	"
T3	"	"	8	5,6,24,3,2,38	27	"	"	"	"
T4	"	"	24	5,6,8,3,2,38	4	"	"	"	"
T5	"	"	3	5,6,8,24,2,38	23	"	"	"	"
T6	"	"	2	5,6,8,24,3,38	22	"	"	"	"

TABLE XX. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Output crosstalk, high level	5	Open	6,8,24,3,2,38	25	0	400	mV	20,21
U2	"	6	"	5,8,24,3,2,38	26	"	"	"	"
U3	"	8	"	5,6,24,3,2,38	27	"	"	"	"
U4	"	24	"	5,6,8,3,2,38	4	"	"	"	"
U5	"	3	"	5,6,8,24,2,38	23	"	"	"	"
U6	"	2	"	5,6,8,24,3,38	22	"	"	"	"

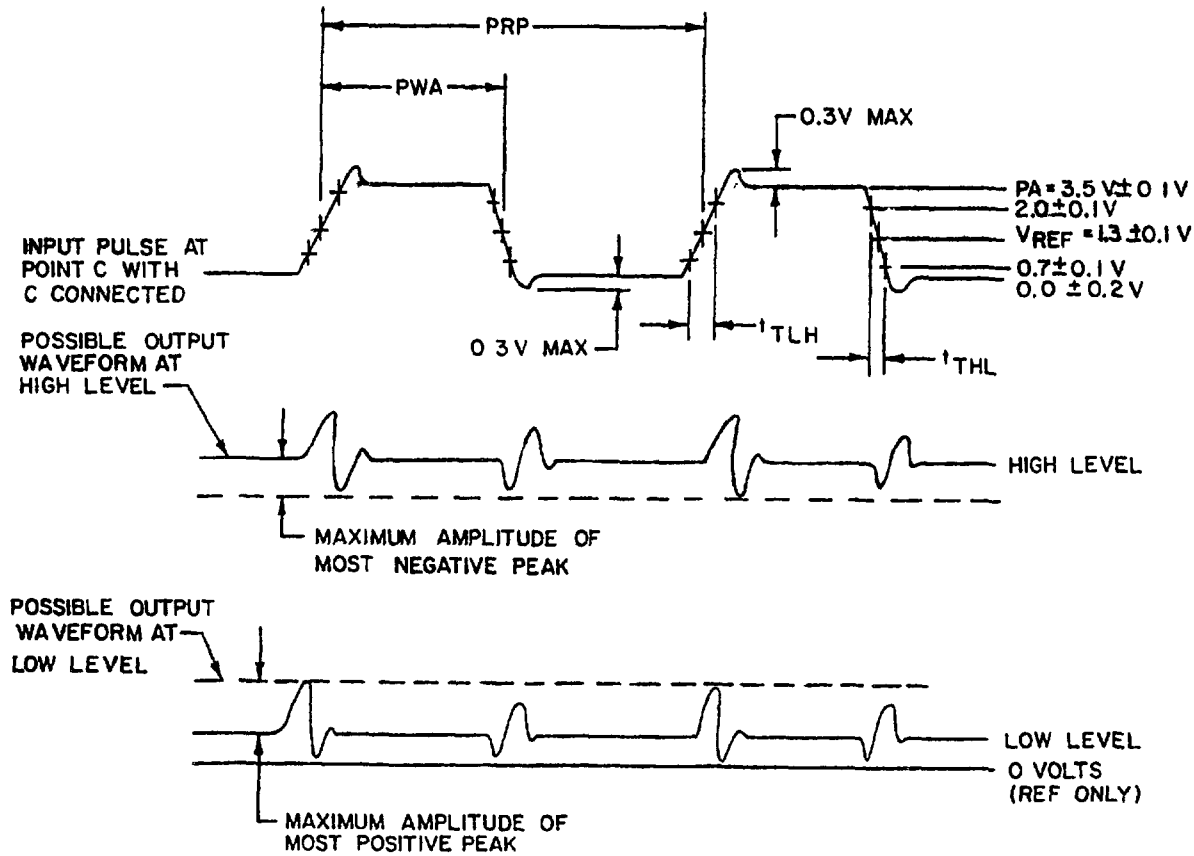
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 20. Output crosstalk (low-level and high-level) test circuit for tables XIX and XX.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

FIGURE 21. Output crosstalk (low-level and high-level) test circuit waveforms for tables XIX and XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE XXI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	400	mV	22,23

TABLE XXII. Power supply filter capacitance test.

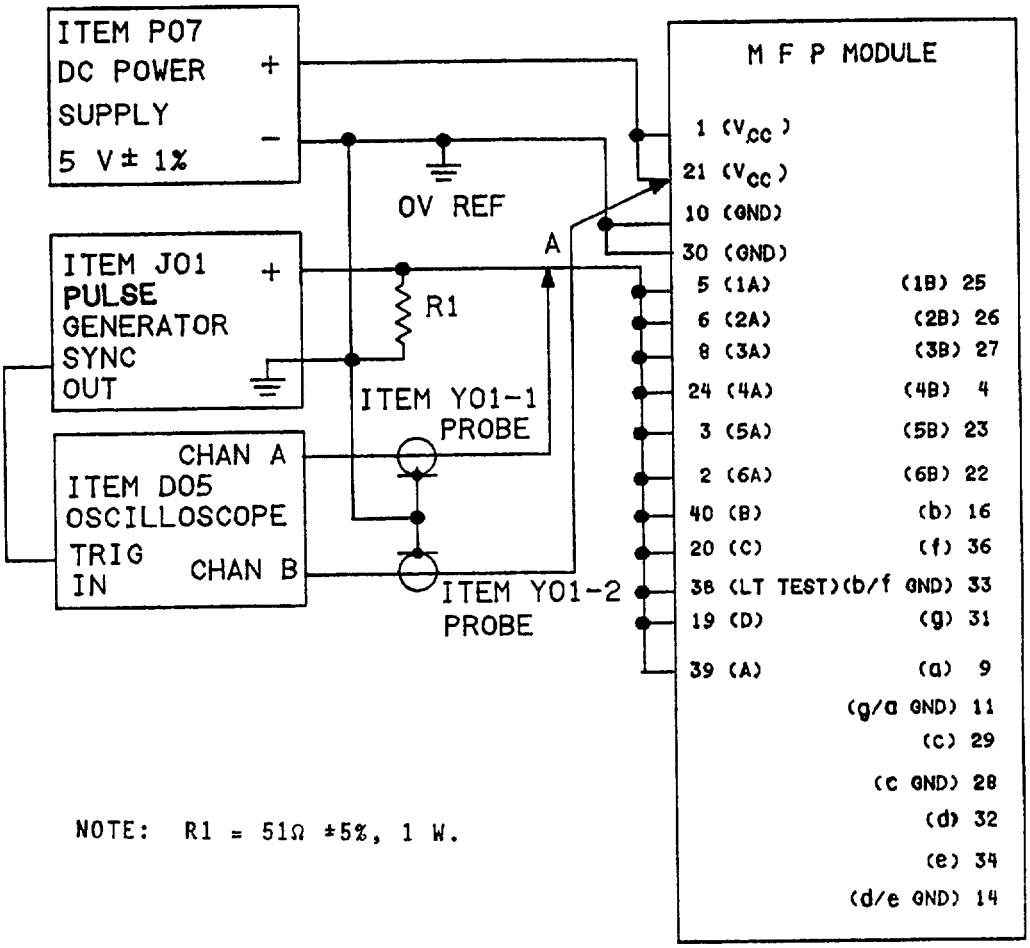
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	Open	1,21	1.6	3.0	µF	24

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	Input capacitance	5	Open	2	30	pF	24
X2	"	6	"	"	"	"	"
X3	"	8	"	"	"	"	"
X4	"	24	"	"	"	"	"
X5	"	3	"	"	"	"	"
X6	"	2	"	"	"	"	"
X7	"	40	"	"	"	"	"
X8	"	20	"	"	"	"	"
X9	"	38	"	"	"	"	"
X10	"	19	"	"	"	"	"
X11	"	39	"	"	"	"	"

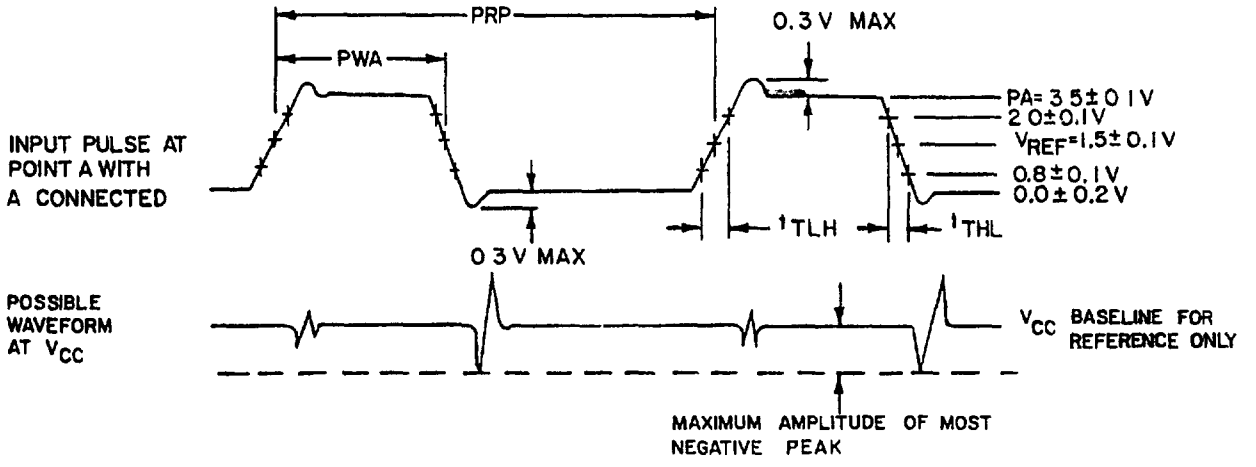
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 22. Power supply transient amplitude test circuit for table XXI.

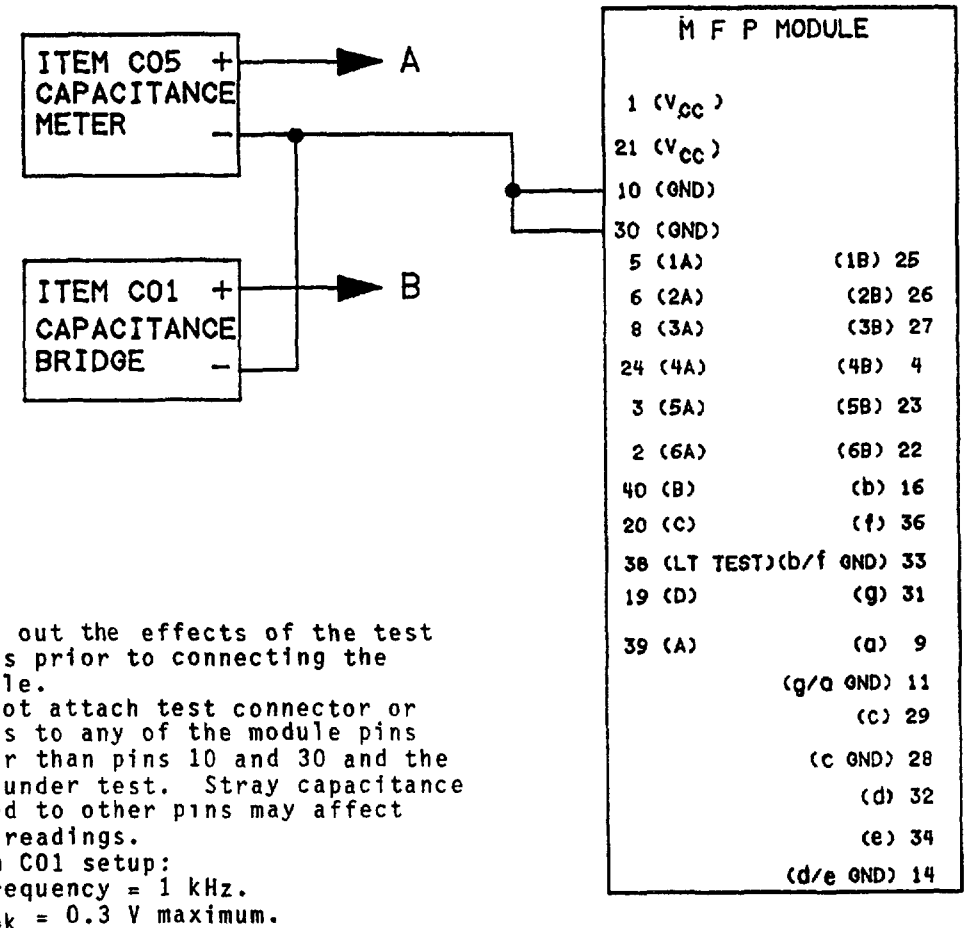
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:49Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PWA = 1,000 ±50 ns.
PRP = 2,000 ±100 ns.

FIGURE 23. Power supply transient amplitude test circuit waveforms for table XXI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 24. Power supply filter and input capacitance test circuit for tables XXII and XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NESC
ATTN: ELEX 8111
Washington DC 20363

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
ATTN: ELEX 8111
Department of the Navy
Washington, DC 20363

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/332(EC)	SEM, Decoder, Digital, Key Code MFP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b Recommended Wording

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 20240-20117:49Z
Check the source to verify that this is the current version before use.